

Matrix functions and stability

Symmetric maximizers for Lyapunov operators of order six

Difficulty: challenging **Importance:** interesting to specialist**Status:** Open

Rating rationale: Challenging because the last undecided dimension sits between proved cases and counterexamples; specialist impact reflects the surviving order-six norm identity rather than the refuted all-orders claim.

Problem statement

For each real matrix $A \in \mathbb{R}^{6 \times 6}$, define $L_A(X) = AX + XA^T$. Write $\|X\|_F = (\text{tr}(X^T X))^{1/2}$. Is it true that

$$\max_{X \in \mathbb{R}^{6 \times 6}, \|X\|_F=1} \|L_A(X)\|_F = \max_{X=X^T, \|X\|_F=1} \|L_A(X)\|_F \quad \text{for every } A \in \mathbb{R}^{6 \times 6}?$$

This concerns the **largest** singular value of the Lyapunov operator, with no stability assumption on A .

Why it matters

Restricting a norm calculation to symmetric matrices substantially reduces the associated singular-value problem. The conjecture asks when that reduction retains the exact operator norm, an issue relevant to matrix-equation norm estimation.

References

1. D. Kressner and B. Vandereycken, *A counterexample to the symmetric-maximizer conjecture for Lyapunov operators*, arXiv:2608.20875v1, 21 August 2026, §§1 and 5. [Primary text](#).
2. R. Byers and S. Nash, *On the singular “vectors” of the Lyapunov operator*, SIAM Journal on Algebraic and Discrete Methods 8(1) (1987), 59–66. [DOI](#).

Status check — 2026-09-08

Reference 1 explicitly leaves order six open, while refuting every order at least seven. The original all-orders assertion must therefore not be catalogued as open. The latest arXiv record is v1. Searches included *symmetric-maximizer Lyapunov order six*, *Lyapunov conjecture 2026 proof*, and the exact paper title; no subsequent resolution of order six was located. This is a targeted literature check, not a certification of openness.

Audit — 2026-09-10

Rechecked the [August 2026 counterexample paper](#), §5: order six alone remains unresolved. Exact-title and order-six searches found no subsequent resolution. The importance rating now reflects this narrow surviving target; the broader symmetric-maximizer assertion has already been refuted.